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Roll No

MI-3004-CBGS

B.E. III Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Rock Mechanics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain Brazilian Test in brief.
b) Describe the procedure for determining the density, porosity and moisture content of rock samples in the lab.
2. a) Write notes on Q system.
b) Explain Bieniawski RMR system of rock mass classification and its application.
3. a) Explain Coulomb-Navier theory of failure of rock.
b) Compare Maxwell model with St. Venant model.
4. a) Which are the responsible factors for mechanical properties of rock masses? Write a note on elasticity and plasticity of rocks.
b) Explain swelling. And describe the Swell Index.

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5. Define following:
 - a) Rock fractures
 - b) Point load index

6. a) Write notes on :
 - i) Hydraulic fracturing
 - ii) LVDTb) How size and shape of sample affects the strength of rock sample?

7. Explain CMRI-ISM geomechanics rock mass rating system. How will you measure the various parameters for the roof rock of an underground coal mine and estimate the rock load for support design.

8. Write notes on: (Any two)
 - a) Time dependent properties of rock
 - b) Tangent and secant Modulus of elasticity
 - c) Importance of Rheology
 - d) Maxwell Model

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